

# Associations of dietary dark-green and deep-yellow vegetables and fruits with cervical intraepithelial neoplasia: modification by smoking.

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Smoking has been positively and fruit and vegetable intake has been negatively associated with cervical cancer, the second most common cancer among women worldwide. However, a lower consumption of fruits and reduced serum carotenoids have been observed among smokers. It is not known whether the smoking effect on the risk of cervical neoplasia is modified by a low intake of fruits and vegetables. The present study examined the combined effects of tobacco smoking and diet using a validated FFQ and serum carotenoid and tocopherol levels on cervical intraepithelial neoplasia grade 3 (CIN3) risk in a hospital-based case-control study conducted in São Paulo, Brazil, between 2003 and 2005. The sample comprised 231 incident, histologically confirmed cases of CIN3 and 453 controls. A low intake ( $\leq 39$  g) of dark-green and deep-yellow vegetables and fruits without tobacco smoking had a lesser effect on CIN3 (OR 1.14; 95 % CI 0.49, 2.65) than among smokers with higher intake ( $\geq 40$  g; OR 1.83; 95 % CI 0.73, 4.62) after adjusting for confounders. The OR for the joint exposure of tobacco smoking and low intake of vegetables and fruits was greater (3.86; 95 % CI 1.74, 8.57; P for trend  $< 0.001$ ) compared with non-smokers with higher intake after adjusting for confounding variables and human papillomavirus status. Similar results were observed for total fruit, serum total carotene (including  $\beta$ -,  $\alpha$ - and  $\gamma$ -carotene) and tocopherols. These findings suggest that the effect of nutritional factors on CIN3 is modified by smoking.